ActInf Textbook Group ~ Cohort 4 ~ Meeting 16 (Chapter 7 part 1)

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all right greetings everyone it is October 10th 1010 23 and we're in our discussion of chapter 7 so last week Ali and I recorded like the short video overviews of chapters nine and 10 just because there was um only a few of us during our cohort for slot and we just wanted to complete those um overviews so now we're in the chapter 7 um discussion so we'll jump over to the chapter 7 page um and before we look at like any of the specific questions like does anybody just want to share um what did they think about chapter 7 what what was there impression or or how would they summarize it to to somebody or what was just exciting to them about chapter 7 well can I start um so please hello Daniel hi yeah go forward go forward yeah so so my um impression of chapter 7 is about uh uh it's many about about the uh discrete uh State and discrete outcome model of the active inference and I found it is particularly uh useful to have a detail understanding of one of the example which is this T's uh example and it is kind of the first uh active inference model that I can get my hand on and try it out and change those uh those parameters or the the matrixes used in the active inference model and see how um how the uh active inference uh actually works awesome I think that's kind of the point um anyone else want to give a thought on seven um yeah I would just second everything that uh Lan said just now um I think it's a great way to actually see maybe how some of the the ideas that we've been reading about prior in the textbook actually start to hang together and play out in an example I think that the the use of discret variables is highly like kind of interpretable I think this is a it comes a little bit more naturally to me to read this chapter versus um you know chapter 8 uh with continuous variables because we get to see like your you know more precise specific things that happen discreet choices and and things that occur and then also just some kind of interesting observations that come out of that with the different um kind of components of the model like uh you know as you get further in the chapter there's a discussion of like here are the interesting things that happen whenever you don't include preferences or have highly uh uniform preferences like there's just pure novelty seeking on the part of the agent who like um like in the team's example they just can't sit still they never come back to the same spot they just it's pure just seeking epistemic value it seems because there's nothing pragmatic to do because there's nothing in particular that they're going for because it's not been hardcoded into the initialized um preferences so so yeah just um giving us that that really big model for active inference here in the discret form in addition seeing the interesting dynamics that can kind of play out emerging properties and so on awesome anyone else want to make a general comment on seven I would I would just Echo that given that I've come from a psychology background and so seeing the exploration exploitation Paradigm just being laid out in that way and they make the comment that you know they believe they've uh come to understand how these two go hand in hand instead of being competitive I thought that was also very very intuitive cool yes yeah the continuous time setting is kind of like having your hand on the rudder in a boat and uh so you're making continuous actions and there's a continuous flow in these paths and the discreete time is like chess or Checkers like time clicks discreetly and the actions are discreet the states are discreet and so that can make it kind of clearer a little bit more cut and dry um cool does anybody want to jump to a specific section of seven or we can look at the questions and just like look at at the upvoted ones or if anybody submitted any and just like look at what's interesting in seven or or somebody can propose a section that that they want to look at I'm yeah I'm mostly interest see in chapter uh uh 7.3 uh decision making and planning as a reference uh I wonder if you know if someone else has preference to talk about the sure well let's yeah let's get to 7.3 basically 7.1 is just one paragraph we can see here 7.2 is just a passive inference model this is just the downstairs section of the discrete time model model we have hidden States changing through time being related to observations at each time point and a prior so it's just the downstairs of figure 4.3 and it's kind of worked out in multiple stages with this music listening example so then we get to 7.3 now we take that downstairs the part that looks like an e um and we add in the action so so this is where we actually get active inference and then what makes it active inference under the free energy principle is that the unified imperative for Action selection is going to be expected free energy so right because there are other models that have um sense making or signal processing on the inbound and then decision- making or action selection or control theory on the outbound like that is in a loose sense active inference but when we're talking about active inference we're talking about active inference and the free energy principle which is what we get when we bring in the variational and the expected free energy okay so what what aspect of 7.3 do you want to explore well first about this figure that you're are showing here um I I have a question about uh the fourth line from the bottom uh you you see third while c and e appear in the panel on the right do you see that sentence where is the e i I couldn't find the E yes so e um is the Habit e represents the prior Distribution on policies so let's just say that we have four affordances up down left right the E Vector is like 0.25 0.25 0.25 0.25 so in that situation e conveys through its inclusion which actions are possible and okay by the value in E that is the prior on policy so if they were all 0.25s and there was four um affordances then that would be like saying those options um in the absence of being sharpened by expected free energy are equally likely to be taken and um we can look at um the the nervous system diagram in figure 5.4 so here's e so here on the indirect pathway observations are coming in habits are being triggered on the direct pathway we have expected free energy sharpening our habit distribution that doesn't just me yeah it doesn't just mean exaggerating it it could upweight something like let's just say there's a rare policy um like but a given situation it has high epistemic Andor pragmatic value well then you'd hope that even though your prior on um doing it is very low like what's the prior on taking off your seat belt well on a 10hour car ride it's low but then when the car is stopped and you want to leave that's an amazingly Salient thing to do so e basically hosts what can be done and in the absence of any decision-making process the distribution of actions to be taken and then G sharpens that to to to to update the policy posterior and then the policy posterior is where action is sampled from I see I see from from the figure 7.3 it seems that c and e plays the same role as f in the distribution of poliy right they kind of Absol R um absolutely like um free energy expected free energy here G can be understood as a statistical distribution about policy conditioned upon CN we we just discussed why e matters for the calculation of policy posterior C matters because of equation 2.6 so C matters for G calculation because C is going to decide what is of pragmatic value so um in that representation that they have in 7.3 basically just saying um policy is being computed conditioned upon preference and on habit okay so C is preference e is habit so they they encode different uh kind of preferences of the policies yeah okay thank you denner that's very very good cool yeah I mean anyone else have like a thought on 7.3 or or a question on 7.3 or just how how they see it or how they interpret it so I I have further question so again on this figure 7.3 so we see the policy Pi affects the um the the um probity uh State probity transition Matrix B uh I wonder uh if it's possible to model the uh policy also affect the Matrix a meaning that it it it not only affects the perp transition mes Matrix but also um affects the transformation from the uh state to the outcome great great question all all kind of mention two ways that you could think about that so as you mentioned policy intervenes in B how things change through time like when you turn on the air conditioner it doesn't like reach into the room and just like modify the thermometer it modifies how the temperature changes through time so it modifies the Dynamics of the Hidden State the temperature and then that results in a thermometer changing so first answer to the question is we don't need policy to change a if we want if we prefer it to be cooler and we take the policy to turn on the air conditioner then at in the next time step the temperature of the room is going to drop let's say one policy is you know turn on the air conditioner temperature drops by one turn off it it goes up by one so you turn it on the temperature drops in the room and then the the thermometer is cooler so one answer is you don't really want to touch a because the mapping between hidden States and observations isn't really something you can act on directly and so we're intervening in the real underlying hidden causal process and then that emits OB the first and the second answer is you could develop an active inference model where a metacognitive agent selects which a matrix to use so um maybe you decide whether to put on sunglasses or not and then when you have sunglasses on colors are mapped to your vision this way and when you do something different they're mapped a different way so that would be taking a policy choice on which a to use okay but in most settings the relationship between hidden States and observations is um it's orthagonal to or you could say it's factorized away from how hidden States change through time keeping this model sparse in the sense that keeping as few edges as possible keeps the comput ational cost low and keeps the um kind of parsimony High let's see thanks than that's very good so um yeah so so it really depends on how states and the outcomes are are defined um maybe in my case we can Define it such way that that high the policy affects only the states transition perity not affecting the um the trans Nation from the state to outcomes yeah in in almost all cases you will have policy only intervening in B so making a generative model I mean this chapter six making a generative model is if you can describe what all of these are in your system of interest you literally have the generative model and how do you categorize different things that you have or want to include into this well observations are literally the data if it's in a spreadsheet it's literally an observation then right policies are actions so if something is something that an agent can do and then everything else is a hidden State now if you're in a fully observable situation like a chessboard then it can seem like there's a redundancy like the observation is the location of the piece and the hidden state is the true location of the piece so in that case it's not really a partially observable Markov decision process pomdp it's just an mdp because it's a fully observable decision process but a allows us to bring in the partial observability but if you're in a fully observable setting then you you can almost think about direct intervention however even in that case then the a matrix is just an identity Matrix like it Maps perfectly between hidden States you know you have an accurate thermometer 37 just maps to 37 then a is an identity Matrix and it has basically no computational cost and then your model is more flexible in case you ever did have a noisy thermometer or a missing measurement if you had a pmdp then it's fine but if you were to only have the mdp then you can't deal with the noisy or a partially observable setting yes thank you thank you Dan that's very detailed exper yeah I'll hold on uh hold my questions for okay other participants to ask yeah anyone want to just give a a thought or or a question on this section um yeah I guess just since it's it's been a bit since I actually read like really did a deep dive on this particular chapter um but um I just I'm trying to make sure I grasp like whenever they're describing the the teamas shortly after um yeah I guess we're still in 7.3 they just start to describe like the a matrices is like in this case they become tensors um like with the teamat you have there are outcomes and then there are locations and then there are contexts so I guess the you know the the mouse uh searching for a reward is looking for I guess the outcome is like whether the the space they land on is a reward or not um and um the location is just you know them navigating the maze itself and the locations of the Maze and then also you have the the context I might have I might be mixing those up a bit but it's just just trying to make sure I can grasp like we're looking at potentially 3D 4D and so on um like kind of tensor um you know quantities here that are based on all of the kind of Dimensions to what what the the mouse is doing rather than just uh yeah looking at as like a 2d thing as As Nice and simple as that would be yes um ba basically yes so a I mean you could think of it as okay well pull back the context is whether you're in um world State a or World State B so the two contexts are whether the the food is on the left or the right um in in in in in this setting you just you did mention that um okay so now a Maps between observations and hidden stat States there's going to be two types of hidden States and so each of them projects out to a given observation A1 is about location A2 is about context I don't know if you would properly call this a tensor but certainly this is two matrices um but it it's not but they're different shapes this is a 5x5 Matrix and this is a a a 3x4 matrix when we get to B it will be a true tensor in the sense that every slice of B is going to be a matrix with the same dimensionality but here it's kind of like two parallel channels like this would be analogous to having two sensory modalities so if if we had had vision and hearing those a matrices um could map to the same hidden State like maybe you know is there lightning um or is is there a thunderstorm and you might have a mapping between hearing and is there a thunderstorm um and vision and is there a thunderstorm or you can have those two modalities maap to two different hidden State modalities like in this case I just want to make one more point before we continue um forward in the teas because um it was brought up about the explore exploit the most probable policies are those that lead to the lowest expected free energy again this is one of the key distinguishing features if not the key distinguishing feature between active inference guided policy select ction and reinforcement learning or reward maximization there is no proposal of a reward or utility function that gets optimized there's just the definition of the generative model and then the generative model does the likeliest thing given its setup again that doesn't mean the animal lives it doesn't mean that XYZ it just means that given how the generative model is set up it does the likeliest thing and then given empirical data chapter nine we set up or parameterize the generative model to be the likeliest one so Daniel can yeah go Ahad sorry yeah I I do have a question regarding 744 where you're right there yeah here uh so I'm I be confused what p and Q means um do we always use P represents the uh prior or the true distribution and Q as a approximate distribution Q is is is this Q the one we actually want uh decision that we uh we intend to um to change our belief by updating uh approximation to the uh real real probability so p is the real probability exactly Q is the real mapping in this case um so it's it's a natural log so this is kind of like taking a surprisal okay it's a it's of observations conditioned upon preferences so this is really what What You observe in relationship to what you prefer you and the reason why it it has the p is because we're talking about like the the actual observations coming in and Q is the variational distribution that we control okay okay yeah now it's clear so actually we can change Q to minimize this GQ GP this object function here right yeah um yeah like Q are about our beliefs whereas we don't really get to control in the same way like we Q of observations conditioned upon policy so what do I believe the sequence of future observations will be ilda conditioned upon what I do that's all in the mind that is different than what is the sequence of observations given my preferences you can change your preferences but that doesn't change observations given your preferences and similarly you could change your belief about hidden States but you can't in the same way control the mapping between observations and hidden States in actuality yeah and this is like a restatement in equation 7.4 of equation 2.6 expected free energy so we have same epic and pragmatic value okay so let's go through the teamas and just see how this is going to play out okay so the two factors so factor is being used to describe a different kind of hidden State um so again there's two factors here there's the context factor and the location factor one of those is a controllable Factor location whereas context is not a controllable Factor if it were there would be another policy type change the food to the left to the right but we're not in that setting so this example brings up the distinction between controllable and uncontrollable factors observations they are the fundamental observables that's in the spreadsheet then everything that's not an observation is a hidden State Factor State factor and then some State factors are controllable those are the ones that have policies intervening in their B matrices other state factors are uncontrollable it's just a latent cause of the world that's proposed within a cognitive model but with respect to that uncontrollable State Factor that's like listening to music the first example that's like a passive inference on that state Factor okay so the controllable state are the are the uh are the locations of the mouse right right because it can move uncontrollable states are the locations of this yeah because the mouse doesn't get to decide where the food is right yeah okay yeah um so that then that's where agency and self-efficacy can come into play is to what extent do actions bear um Can can the agent take actions and make a difference that makes a difference take an action that changes some state in the world controllable factor and how are and then this is all about how they evaluate that it could be chosen because it has high pragmatic value because doing that kind of exerting that kind of control brings observations closer to alignment with preferences pragmatic value or it might lead to new information about the world epistemic value and that's the entropy the expected difference of the entropies of observations conditioned upon hidden States in comparison with observations conditioned upon policy so this is basically like the entropy of the a matrix so that's that's the mapping between observations in Hidden States here we're asking how much is it going to matter differently if we condition upon policy so let's just say that the a matrix has a a given entropy the a matrix has a given uncertainty associated with it the least uncertain a matrix is like the identity Matrix just one on the diagonal and all zeros elsewhere you have perfect knowledge it becomes a fully observable setting the most entropy a matrix is just All Uniform basically the measurements don't even matter because all the measurements are equally likely to map to any hidden state so that's on the left side and now the question is conditioned upon action is there going to be a difference in the sharpness of that mapping well if it's a policy that doesn't lead to sharpening of the a it doesn't have episte IC value in comparison if there's a policy such that um you believe that taking that policy is going to sharpen your understanding of outcomes then this is a non-zero value so that's the epistemic value yes Daniel the um however you said that you know from the model that a matrix is is fixed right it's not affect by you know all the entries in a matrix are not affected by the PCY Pi because PCY Pi affects only the state transition probability Matrix B and what do you mean by sharpen a using policy Pi good question so there's there's a few um a few pieces here first off there's a matrix learning that is not happening in this example so okay but but but but but it is it is a cognitive phenomena that the a matrix could be updated but it's not being there's no learning here um now let's just say that there's um we have we have a a two observ we have observations and and hidden States one of the observations doesn't lead to gain information about hidden States like it Maps 50% of the time to A or B another observation gives you a lot of information uh information about the world then that policy can the policy that that leads you to get that OB the informative observation that has a high um epistemic value now that doesn't mean that policy reaches into a in the factorization of the base graph right so it's it's not like every single thing it's like saying well um how could what the finger does influence knee they're not connected directly it's like well yeah but the whole generative model is executed as a whole so um you can I mean to put it a different way you can use expressions in the active ontology that associate different variables that aren't necessarily connected through an edge here so these edges are not the only possible sentences that you can construct when talking about the generative model so you can talk about how G is related to O right OB obviously because observations are what are o Tilda are what are being taken into account when we're talking about G but there's no line between G and observations so this is just asking um will those observations be informative and and the observations being informative doesn't need to entail update of the a matrix I see I see yeah good point good yeah or you know there there's two people you can ask one of them he flips a coin the other one right tells you the truth so the epistemic value with going to the person who gives you the good information is high even if you have a fixed understanding of that situation okay let's look at it in the team A's setting yeah yeah okay okay so so technically okay technically the a matrices are tensors but it's a little bit of an irregular one okay so here we have these two kind of parallel A's on the the first and the second [Music] um uh context of a or first and the second factors mapping um the actual location here on the rows note that there's this L and the r this is because the hidden states are in the cases of being on the right being on the left and being in the starting position those are all identity mappings but now because we're in the black on the right setting the epistemic Q at the bottom when you are physically in the bottom you go to the hidden state which tells you you're in the r context and there's all zeros here that hidden State cannot be reached but of course when the context is Switched it's the exact opposite there's a one here going to the bottom Maps you to the it's the L State and there is no way to reach the r State and we also see a flipping of the 98s 98 is here on the bottom left and on the right and here it's on the top left on the right this is again just just to show these matrices are necessary and sufficient for specifying the situation they describe literally exactly what is happening all that your generative model knows is what you specify in these matrices there's no other information so that's the a matrix this these this is showing the two contexts and it's showing um the mappings between um and this kind of like location plus distinction between the messages that the epistemic Q on the bottom could provide and then also notice that um when you're in the starting or the bottom position you're you're in the sort of like dashed line like you're not finding out anything but then when you're in but then there's a 98% chance you have a 98% sensory accuracy there still is a 2% mapping between um you know um if you know if you're on the left in this context and it's black there's there's still a 2% chance it maps to White okay now we get to B so similarly we're going to have two B matrices um or two B tensors for each of these State factors B2 they're going to introduce first because B2 the context stays constant over time and can be represented as an identity Matrix so basically this is just keeping the same thing through time within a trial and this is where we get the the true B tensor here this describes changes as a function of movement selection so we have from and to and then there's four actions B does all the work let's look at the generative model like observations are just data points hidden states are hidden States um they're they're just descriptors proposed or actual Laten descriptors and then this is just a sensory mapping but all the work of the meaning of embedded action is going to play out through b the dimensionality of B is also higher because B is whatever the dimensionality is of the state Factor then as many slices as there are policies yeah because each policy is going to change how the world changes now if all of these slices were the same that would be like saying whichever policy you select the world is going to change in the same way so in that situation there's no real causal efficacy or agency of that agent yes so B is really where it's at and and in this example the B is just given and fixed but there are other examples that we could explore that you can look up where there's B Matrix learning so that's learning the consequences of one's actions through taking different policies so that's a situation kind of like a a multi-armed band it where you actually have to learn the mapping between different things preferences here C2 is the um the context Associated preference Vector so we have uh if there's new the dashed line is neutral so this is these three elements corresp respond to these three elements 0 6 negative and six and then these five in the top C correspond to these five um and then as they say there's there's a slight they give a negative one to the starting position just to kind of bump it off and get it going because if it really had no preferences it it I mean all of this comes into the parameterization why is it six and not nine and negative 9 why not nine and -2 that is the parameterization question so when we're in a pedagogical setting and we're just like looking to play around that's when it's super helpful to have the the code and like do parameter sweeps or just by hand play around with how different variables change the outcome of the model um and then in chapter nine it discusses more well how would you go from a bunch of videos of rats in the Maze to inferring what the likeliest parameters are but here they're just given um also just note that these values are given in um a way that it's actually like e to blank so the six here has um actually a a interpretable meaning in terms of the ratio of the preference so this is not just like six arbitrary points or like I don't know it is just literally describing kind of like the the odds ratio right d in the context um domain it's just one2 one one or it could have just said 0. 5.5 but they're they're conveying that it has to sum to one something must happen this is a distribution and the values are the same so uniform distribution this is a perfect knowledge over the starting position right Daniel can ask a question here so this D2 is half and half perity right but this is not known to the agent who makes the active inference right it is known it it starts out completely unsure it starts out with a 50-50 belief but it's kind of funny because it's like it's 100% sure that it's 50/50 like same as a coin flip you're 100% 50/50 if you had a more sophisticated model and and it is shown in a later figure you could have a hyper prior on D so you could ask how confident am I about the prior you can do that for any of them okay okay so you have a hyper prior that can uh characterize D2 uh maybe character as some parameter right so you can later on learn learn those exactly exactly any parameter that you want to learn in a bayian statistic setting you need a hyper prior if you don't if have if it's just a fixed parameter then it's like the equivalent of the derac Delta function it's just like a spike like you have like it's just like a single observation right yeah um and and it's always a very interesting thing in this case or in in your own generative models like what is being made explicit and then what is actually being kept implicit for example like how does the rat know that it's going to get the information at that location right more more question how does the r know it's half and a half it's just given it's just given I mean that's like asking how how does the rat know that moving um to the left is going to get it to the left but but also um keep the uh figure figure 9.2 in mind weird it's like it can't 9.1 we're making a map of the Rat here is literally figure four figure 4.3 or figure 7.3 right there D is not shown so we're just describing its map of its space we're not claiming to make a total account of everything that the rat knows um but yeah I mean we there's more to say on this example then there's an i example here's here in 710 here's the prior on D here's the lowercase letters or hyper everything then another example with a maze finding agent and then a hierarchical nested model where you you put a GM as a leaf in another GM and that is shown in a nested example so it's kind of a long chapter actually but there because there's a lot of examples that are kind of like densely packed but just in our last like minutes here what what does anyone find interesting or what questions they still have well um I still have one more question in in the in the end of section 7.3 okay um uh so it's uh two sentence above 7.4 okay yeah you're still at uh okay okay um for I think further up um so it talks about the neuron neuronal activity uh okay yeah so the second to the last sentence uh it talk about field potentials and neural activities right uh in figure 7.2 yeah um so so my question is how realistic is this um Chim model uh comparing to the experimental data so we have this is ass simulation right so basically we constructed active uh inference model using the data to simulate the activities of the of of a mouse in in the tmat uh so my question is how does the simulation results compare to the uh actual biological experiment the activities of the mouse and how they um the brain the mouse brain um uh activities you know uh imply or or corresponds to this numerical simulation I'm not aware of them doing this in a rat doing the te- but it may exist we should look it up in other settings it Maps very very well now that's just like from from my perspective it's just a bonus because active inference is not developed as an account of brains the particular partition the basian mechanics all of the cognitive physics doesn't need to be about any specific system however as the empirical lines of work have shown various features of the model that are non-trivial are recapitulated in Biometrics that suggests that the analytical solution and the biological embodiment have a convergence strategy an example of that is predictive coding predictive coding was invented by humans as a way to do video encoding because it turns out that frame differencing is the most efficient way to do video encoding like instead of just trying to compress every frame you basically compress the B Matrix because if something doesn't change you don't even need to mention it in the next slide so right predictive coding was a purely information theoretic consideration but it turned out to be utilized in neural coding situations so I see it as positive where it aligns but um where it doesn't align then it it it it ALS it doesn't it just means that there's something else happening that that isn't just narrowly that yeah because you see in figure 7.2 they show the neural activities and the fireing Rays and local field potentials so there must be some mechanism to translate the outcome of the numerical simulation into those neuro activities right the the biological uh uh embodiment of the numerical size um so through what mechanism do they uh trans transform this numerical into into the uh into the field potentials for example neuro activities yeah it's all in the mat lab code in the mat laab functions that are used it describes it in more detail but basically the lfps are associated with with belief updating how about fire race are they associal with yeah decision you could imagine different different firing rates or different different I mean and this has been studied extensively in the fmri and EEG setting um you know does does the um Erp in the ERG or does the local field potential associate with a more intense stimuli with a more expected stimuli with the less expected stimuli with learning those are all hypotheses that people have explored and for different stimuli in different settings you see different patterns okay any last um comments here um I mean I I don't want to keep us too long but I just yeah I still as the last bit I find the Novelties seeking uh rather interesting like just the example it's roughly around page 145 146 like if you were to model give it like if there were an absence of preferences for an agent then like I'm just quoting here the drive to novelty resolution given by expected free energy minimization leads our simulated creature to avoid any previously visited locations um just like you know it becomes like a purely like the model is attempting to uh resolve expected free energy if I understand this correctly by just it's just seeking more and more information it's going to places it hasn't been before um there's nothing pragmatic about returning to previous places it's not attempting to realize any preferences that say I need to go back right so I just thought that's a really nice kind of kind of emergent property that we see whenever you leave out preferences here and it it for me it it kind of lays a lot of emphasis on like the importance of of preferences they're impact upon behavior in these models and just maybe makes me think of like how to you know maybe start with preferences how to model those appropriately because it can highly impact um right like an agent's Behavior yep great points and the strength of the preferences overall and and where that ends up on a Continuum between novelty seeking and basically ruthless pragmatic indeed all right thank you all um we go into chapter eight next time in the later time slot so peace till next time bye thank you Daniel Day